

Carbon Lifecycle Integrity™ Standard (CLI™)

Independent Methodological Authority

OOF™ Origin Open Foundation™

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Canonical Language: English

A. Standard Abstract

The **Carbon Lifecycle Integrity™ Standard (CLI™)** establishes a structured methodology for transparent carbon visibility across the lifecycle of a product.

Instead of treating carbon emissions as isolated operational values, CLI™ defines carbon exposure as a **lifecycle property emerging from the full operational chain of a product.**

The standard introduces structural conditions that allow organizations to map, disclose, and evaluate carbon exposure across material origin, production processes, packaging systems, and distribution pathways.

Carbon Lifecycle Integrity™ does not regulate emissions, calculate mandatory values, or certify carbon neutrality.

It establishes the structural transparency required to understand **how carbon is generated across a product lifecycle.**

B. Canonical Definition

Carbon Lifecycle Integrity™ (CLI™) is a methodological standard establishing structured lifecycle carbon visibility through boundary-defined mapping of material origin, production energy use, packaging impact, and distribution pathways associated with a product.

Carbon exposure becomes structurally visible when lifecycle boundaries are declared and all carbon-relevant operational stages are transparently

documented.

CLI™ defines the **architecture of lifecycle carbon transparency**, not emission limits or reduction targets.

C. Purpose

The purpose of Carbon Lifecycle Integrity™ is to:

- establish transparent lifecycle carbon visibility for products
- connect material origin, production processes, and logistics into a unified carbon context
- enable consistent carbon-per-product evaluation frameworks
- create structural readiness for future lifecycle carbon reporting systems

CLI™ transforms carbon information from fragmented operational data into a **coherent lifecycle structure**.

D. Structural Scope

Carbon Lifecycle Integrity™ applies to lifecycle stages including:

- raw material origin and extraction
- material processing and preparation
- manufacturing and production operations
- packaging systems and packaging materials
- logistics and distribution pathways
- delivery to a declared commercial or consumer boundary

Lifecycle boundaries must be explicitly declared to maintain structural validity.

Optional lifecycle extensions may include post-use or recycling phases when relevant.

E. Core Structural Requirements

A product may be considered structurally aligned with Carbon Lifecycle Integrity™ only when the following conditions are satisfied:

1. Lifecycle boundaries are clearly defined
2. Material origin and primary processing stages are documented
3. Production energy sources are declared
4. Packaging materials and packaging structure are disclosed
5. Transport modes and distribution pathways are identified
6. A responsible entity publicly acknowledges accountability for lifecycle carbon transparency

Incomplete lifecycle mapping weakens structural integrity of carbon visibility.

F. Lifecycle Transparency Principle

Carbon Lifecycle Integrity™ is based on a simple principle:

Carbon becomes measurable when the lifecycle becomes visible.

When the origin of materials, energy inputs, production stages, and distribution routes are documented, carbon exposure emerges as a traceable structural property of the product.

Transparency enables evaluation.
Evaluation enables improvement.

G. Methodological Position

Carbon Lifecycle Integrity™ functions as:

- a lifecycle transparency framework for carbon exposure
- a compatibility layer between supply chain transparency and environmental disclosure
- a structural preparation layer for product-level carbon evaluation systems

CLI™ does not replace regulatory carbon accounting systems.
It creates structural clarity that allows such systems to operate effectively.

H. Non-Function Clause

Carbon Lifecycle Integrity™ does not constitute:

- carbon neutrality certification
- carbon offset validation
- emission reduction verification
- taxation enforcement mechanism
- regulatory compliance mandate

CLI™ establishes **methodological transparency**, not regulatory authority.

I. Strategic Significance

In modern supply chains, carbon exposure is often hidden within fragmented operational stages.

When lifecycle origin, energy inputs, packaging systems, and logistics pathways are mapped within a unified structure, carbon exposure becomes structurally observable.

This enables:

- product-level environmental transparency
- supply chain optimization
- meaningful comparison between production models
- long-term environmental accountability

Carbon Lifecycle Integrity™ transforms carbon discussion from abstract reporting into **visible product lifecycle architecture**.

Canonical Closing Statement

Carbon exposure does not originate at a single point.

It emerges from the full chain of material origin, production activity, packaging systems, and distribution pathways.

Carbon Lifecycle Integrity™ establishes the structural transparency required to see that chain.

Related Documents

[→ About Carbon Lifecycle Integrity™ Standard](#)

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